

Orbital Propagation and Uncertainty Quantification

A 2024–2026 Literature Review

MARKDOWN · ~150 PUBLICATIONS, 12 AREAS

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§ Abstract

The most promising advances in orbital propagation converge on three themes: adaptive-order Taylor integration reaching production maturity, differential algebra methods proving their operational viability, and an unexploited gap in operator splitting for satellite dynamics. This review spans 12 technical areas across ~150 recent publications and identifies several combinations and overlooked techniques that could genuinely exceed the state of the art.

The most striking finding is that operator splitting applied to artificial satellite dynamics — despite its natural fit to the near-integrable Kepler-plus-perturbation structure — remains essentially unexplored, while Koopman spectral methods, often assumed niche, have exploded into a 15–20 paper subfield in the last two years alone.

CONTENTS

- 01** Taylor integrators (heyoka, Tan/Smith/Rackauckas adaptive order)
- 02** Modified Chebyshev-Picard Iteration (Junkins, Woollands, Bai)
- 03** Differential Algebra and Jet Transport (Massari, Wittig, Armellin, Di Lizia)
- 04** Koopman operator methods for orbital mechanics
- 05** Operator splitting and exponential integrators (the unexplored gap)
- 06** Neural ODE / differentiable propagators (dSGP4, ThermoNET, ML- ∂ SGP4)
- 07** Scientific foundation models and their limitations (Vafa et al., Liu et al.)
- 08** Uncertainty propagation beyond Gaussian (GBEES, particle flow, normalizing flows)
- 09** Symplectic and geometric integrators
- 10** GPU-accelerated propagation (jaxsgp4)
- 11** Geometric / Clifford algebra applications
- 12** The Mojo programming language ecosystem for astrodynamics

§ Taylor integrators have crossed the operational threshold

The heyoka ecosystem (Biscani & Izzo) has evolved from a research-grade Taylor integrator into an operational Earth-orbit propagation framework. As of `heyoka.py v7.10.1`, it supports EGM2008 geopotential, NRLMSISE-00 and JB-08 atmospheric models, IAU2000/2006 precession-nutation, Earth Orientation Parameters, ICRS/ITRS/TEME frame transformations, and VSOP2013/ELP2000 ephemerides. The LLVM JIT compilation pipeline exploits AVX-512/Neon SIMD, and batch mode processes multiple initial conditions simultaneously with 2–8× SIMD speedup. High-order variational equations — computing arbitrary-order flow derivatives while exploiting Schwarz symmetries — are built in, and Neural ODE support enables embedding learned atmospheric models (ThermoNET) directly in the equations of motion.

The February 2026 paper by Tan, Smith, and Rackauckas (arXiv:2602.04086) introduces the first comprehensive adaptive time-and-order Taylor algorithm for general ODEs. Using Julia's multiple dispatch and TaylorDiff.jl, they generate optimized recursive code via symbolic-numeric computing — no user-facing macros or code rewriting required, unlike

`TaylorIntegration.jl`'s `@taylorize`.

§ Proposed advances identified

- **Operator splitting for satellite dynamics** as a genuine blind spot despite mature mathematical infrastructure from planetary dynamics.
- **A semiglobal Chebyshev-in-time × Taylor-in-state polynomial propagator** combining MCPI and differential algebra — unrealized despite both components being mature.
- **Koopman–exponential integrator fusion** as an unexplored combination.
- **GBEES on GPU** for full-PDF uncertainty propagation enabling real-time non-Gaussian filtering.
- **Normalizing flows for orbital uncertainty** as an open frontier where the natural geometry (e.g., on the orbital element manifold) has not yet been exploited.



§ Status & provenance

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